

PICOFLOW · DESIGN DECISIONS REGISTER

PicoFlow design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-01	Amish Chadha	Open decision 2 (inlet valve bore, N6) decided by Amish as recommended and moved to Decisions made; open items renumbered; clamp resistor added as a proposed item; value engineering updated for the full-bore valve
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from the review notes, PCF-DDR-001 to 003 and the build plan work

PicoFlow design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design-for-construction changes (frame, tie rods, flange bearing units, runner hub, nozzle saddles, manifold layout, pipe stands, fixings; the valve fittings of P11 are superseded by the full-bore valve decided on 2026-10-01)	Accept; accept with changes	Accept	The whole build plan	PCF-DDR-003, A1
2	Frame made by welding or bolting	(a) welded by a local fabricator; (b) bolted with corner gussets	(a)	Frame (section 3.1)	PCF-DDR-003, A3
3	First site type, region and co-design partner, including confirming that target sites carry about 5 to 15 L/s	Region and partner to be chosen	None yet	Site work: forebay, penstock length, pad, equipment post position	PCF-DDR-001 and 002, O1
4	Split housing for runner inspection (shown in the product renders)	(a) one-piece pipe, as modelled; (b) split into bolted halves	(a)	Housing	REVIEW 2026-09-26, item 1
5	Clear inspection window in the housing (shown in the product renders)	(a) none, as modelled; (b) optional bolted polycarbonate window	(b) as an option	Housing	REVIEW 2026-09-26, item 2
6	Guard material (the renders show a clear guard)	(a) 160 mm PVC pipe, as modelled; (b) clear polycarbonate tube of the same size	(b) if a clear tube is found at that size, so the spider can be checked without removing it	Guard (section 3.9)	REVIEW 2026-09-26, item 3; PCF-DDR-003, P9
7	Render layout choices: rectifier box shown loose beside the unit; tailwater drawn 20 mm below the channel edge	Accept as render choices; change the renders	Accept	Renders only	REVIEW 2026-09-26, items 5 and 6
8	Clamp resistor (R8). With the full-bore valve the worst case (3.0 m, 15 L/s) gives more power, and the 8.2 Ω clamp resistor now settles at 40.1 V, 0.1 V above the clamp's 40 V release point: the bus stays below 48 V, but the clamp stays switched in instead of cycling	(a) keep the 8.2 Ω , 300 W resistor; (b) a 6.8 Ω resistor, which settles at 36.7 V but takes 339 W at 48 V, so it needs a 350 W or larger rating	(b), so the clamp keeps a margin below its release point if the generator constants come out less favorable; the price change is small and is to be quoted	Resistor guard and wiring (build plan section 3.16)	PCF-CAL-001 v0.4, section 6

Item 4 of the 2026-09-26 render review (nozzle holes in the housing) is now part of the design (PCF-DDR-003, P5) and needs no separate decision.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	Generator: rpm per volt, losses and resistance at 200 to 400 rpm, overspeed rating at about 800 rpm, face screw circle and spigot, and a shaft at least 50 mm long	The calculation assumes 10 rpm/V; the plate holes and coupling position follow the face and shaft	PCF-CAL-001; PCF-DDR-003, P7
2	Flange bearing units: 86 mm square, 64 mm bolt pitch, about 33 mm tall, 12.8 kN load rating	They set the lid holes, sleeve length and coupling height	PCF-DDR-003, P3
3	Socket depths and centre-to-mouth sizes of the tee, reducer and elbows	The five pipe cut lengths assume 45 mm sockets and 70 mm elbow centre to mouth	PCF-DDR-003, P10
4	The 90 mm flexible couplings fit the 90 mm printed spigot snugly	The spigot is printed; its outside may need sanding or a wrap of tape	PCF-DDR-003, P5
5	Drainage pipe joints are good for 50 kPa	The surge check (R17) assumes it	PCF-CAL-001, section 9
6	Full-bore 125 mm gate valve: socket depth (70 mm assumed), length over the sockets (330 mm), bore about 117 mm, mass (about 5.5 kg), price (USD 115 assumed), and that it takes about ten turns to close	Sets the pipe piece length, the valve loss and the pipework mass, and enforces the 10 s closure (R17)	PCF-CAL-001 v0.4; PCF-DDR-001, D8; decision of 2026-10-01
7	Forebay tub size and wall stiffness for a 125 mm tank connector	The outlet hole and screen frame follow the tub	PCF-DDR-003, P13

■ Value engineering

Value-engineering target: USD 450 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 614 for the turbine kit with a new generator (USD 164 over the target); USD 544 with a salvaged washing machine motor (USD 94 over the target). The penstock (about USD 140 for 20 m) and the battery (about USD 160) are outside the kit, as decided with the budget. Main cost drivers and savings worth trying:

- The largest lines are the full-bore inlet valve (USD 120, an estimate), the generator (USD 110), the nozzle manifold (USD 50), the controller (USD 43), the bearing units, plate, posts and guard (USD 43) and the housing and lid (USD 35).
- Making the design constructable added USD 86: fixings (USD 28), pipe stands (USD 15), flexible couplings and the reducer (USD 12), flange bearing units and plate (USD 10), valve fittings (USD 8, since replaced by the full-bore valve), the forebay tank connector and screen frame (USD 8), and smaller items (USD 5).
- Savings worth trying: a salvaged washing machine motor (about USD 70 less, already the documented variant); a 6 mm steel generator plate instead of 8 mm aluminium (a few dollars, about 0.6 kg heavier); timber pipe stands; buying fixings as one bulk pack. The full-bore valve (decided 2026-10-01) raised line 8 from USD 40 to USD 120 to recover 3.9 W at the design point, about USD 21 per watt; no published price was found for a 125 mm PVC-U gate valve, so getting real quotes is the first saving to try, and a cheaper full-bore valve of the same kind would bring most of the USD 80 back.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D10: budget USD 450 with the salvaged-motor variant documented, new low-speed BLDC generator, open-design controller (off-the-shelf only for first bench tests), 12 V battery, PETG then glass-filled nylon runner, vertical shaft, two jets, drainage pipe with a slow-closing valve, problem wording, Turgo	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	PCF-DDR-001
2026-09-25	TRL 3 items N1 to N5: 125 mm design-point penstock, R4 relaxed to 20 W, R2 stated per head, R14 turbine-unit definition, USD 2 cost margin accepted	Amish: "i accept all your recommendations, go with them across all repos."	PCF-DDR-002
2026-09-30	Make the design physically buildable while drawing the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	PCF-DDR-003 (changes open for review, open decision 1)
2026-09-30	Outstanding decisions are kept in this register, not in the build plan	Amish: "don't log outstanding decisions in this build plan - that is not the place for it. that should be in a separate design document logged and named as such"	This register
2026-10-01	Inlet valve bore (N6, open decision 2 in register v0.1): option (a), a full-bore valve matched to the 125 mm penstock, re-priced. A 125 mm PVC-U gate valve with solvent-weld sockets replaces the 90 mm valve, reducer and expander: 85.6 W at exactly 10 L/s, 5.6 W over R3; line 8 USD 120	Amish: "picoflow - i agree with the recommendation"	PCF-DDR-003, A2; PCF-CAL-001 v0.4
2026-10-01	The budget is a value-engineering target, not a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	This register, Value engineering